

Real-time Feasible Usage of Radial Basis Functions for Representing Unstructured Environments in Optimal Ship Control *

Trym Tengesdal¹, Sébastien Gros¹ and Tor A. Johansen¹.

Abstract— Navigating the highly unstructured maritime environment poses significant challenges for autonomous ships, particularly in the vicinity of static obstacles and inland waterways. Real-time algorithms capable of comprehending the complexity of nearby grounding hazards are crucial for safe and efficient operations. Electronic Navigational Charts (ENCs) serve as the standard for representing the static environment, providing detailed polygon-based information about obstacles and grounding hazards. However, the non-convex nature of these polygons presents difficulties for optimal control and Model Predictive Control (MPC) schemes. This article introduces an accurate and computationally feasible method for extracting and representing non-convex static obstacle polygons, specifically tailored for use in numerical optimization-based planning, path-following, and trajectory tracking close to static hazards. By employing geometric surface interpolation techniques, Thin Plate Spline (TPS) type of Radial Basis Function (RBF) interpolants are derived from each grounding hazard polygon's point set. These surface approximation functions are then utilized as inequality constraints within a Nonlinear MPC (NMPC) planning scheme to facilitate trajectory tracking with anti-grounding functionality. The integration of surface functions enables the NMPC to consider higher-detail maps, enabling the proposed control system to closely follow trajectories within the unstructured environment's hazardous areas, demonstrated through proof-of-concept simulations.

I. INTRODUCTION

Autonomous ships have a large potential for increasing safety and efficiency in maritime transportation and operations. However, numerous challenges still have to be solved for their potential to be fully realized. This includes the development of efficient and robust planning and tracking algorithms for navigating through the *unstructured environment* the ships are to operate in. By unstructured we mean complex and uncertain environments that are not completely pre-determined. The maritime environment is in general composed of grounding hazards both below and above the water surface. Digital maps, such as Electronic Navigational Charts (ENCs), provide a priori information on the mapped part of this environment with an accuracy dependent on the survey coverage and methodology. The map data is often parameterized as non-convex polygons for the land, shore, and seabed geometries at different depth levels [1]. This makes it hard to employ such data directly into e.g. Model

Predictive Control (MPC) strategies for anti-grounding or static obstacle avoidance, because of the inherent non-convexity, non-smoothness, large memory requirements for considering the map data, not to mention the computational demand of finding the optimal solution when having to consider these grounding hazards with reasonable accuracy. Finding efficient ways of taking into account information on the unstructured environment will allow for less conservative autonomous navigation and e.g. surveys close to grounding hazards.

Existing methods for taking into account information on the unstructured environment directly in the planning involve sampling-based methods such as probabilistic roadmap methods [2] and Rapidly-exploring Random Trees [3], [4]. These allow for the generation of kinodynamically feasible trajectories by sampling and connecting configurations in the collision-free state space from start to goal. Although they can find an initial feasible trajectory very fast, the algorithms can spend excessive time refining the trajectory and thus converge slowly towards the optimum, if at all. Furthermore, it is not yet clear how to robustly and efficiently sample new configurations that are collision-free and ensure quick convergence of the algorithm [5]. Another caveat with such sampling-based methods is that the produced trajectories or paths often need to be smoothed to remove chattering and make the plan suitable for a lower-level motion controller to track. Scenario-based MPC (SB-MPC) has also been applied to the problem [1], where anti-grounding and collision avoidance objectives are posed in the cost function as opposed to being constraints in the MPC. The SB-MPC directly evaluates the finite set of candidate solutions and selects the one yielding minimum cost, not requiring convexity or gradients in the solution procedure. The performance is, however, highly dependent on a rich enough set of candidate solutions. In a similar fashion, evolutionary algorithms such as particle swarm optimization [6] and genetic algorithms [7] do not require convexity or gradients, also being flexible in their formulations and usage. A downside with particle swarm optimization and genetic algorithms is, however, the tendency to prematurely converge to suboptimal solutions and the difficulty in selecting an initial population to improve upon.

The other alternative, as considered in this article, is to employ complete path methods [3] such as optimal control [8]. Complete path methods consider the full continuous state and generate a parameterized trajectory that can explicitly take the vehicle dynamics and kinematics into account. As such, smoothing is most often not necessary. Mathematical

*This work was supported by the Research Council of Norway through the Autoship Centre for Research-based Innovation, project number 309230, and by Kongsberg Maritime through the University Technology Center on Ship Performance and Cyber Physical Systems.

¹The authors are with the Department of Engineering Cybernetics, Norwegian University of Science and Technology (NTNU), Trondheim, Norway (e-mail: {trym.tengesdal, sebastien.gros, tor.arne.johansen}@ntnu.no)

optimization is a central component here, for which analytical solutions are hard to obtain unless simple problems are considered, such as in [9]. As a result, the state and input space are typically discretized in time using a suitable method such as direct single shooting, direct multiple shooting or collocation [10], where numerical optimization is used to find a solution for the state and input trajectories. A popular approach under numerical optimal control therein is MPC, due to its capabilities in handling state, input, and path constraints. Here, the non-convexity of map polygons poses a challenge for which few effective approaches exist. Mixed integer optimization methods as in [2] can be used, at the cost of larger computational demand. For numerical optimal control applied to maritime planning, path-following, or trajectory tracking, a typical approach is to simplify static obstacles as e.g. circular or ellipsoidal representations [11], and unions thereof. An alternative is to approximate an inner convex safe set with respect to obstacles, in which the vessel should stay within [12]. The circular and ellipsoidal approximations can, however, yield very conservative approximations of the environment, in the worst case not allowing for planning or tracking trajectories near land. Furthermore, the inner convex safe set approximation can break down in cases with many small grounding hazards or when planning trajectories "around the corner", leading to an overly small safe set in which it can be difficult to plan or track trajectories over longer horizons [13].

The solution in [14] utilized constrained Delaunay triangulation for representing the collision-free part of the environment, with graph search to find the sequence of triangles leading from start to goal. Numerical optimization is used to iteratively plan a continuous trajectory through the sequence. It can consider the unstructured environment natively but at the cost of a larger computational effort.

In this research, mapped grounding hazards in the unstructured maritime environment are approximated as smooth surface functions, through interpolation of non-convex polygon point sets. This allows for a computationally efficient way of considering information on the unstructured environment when computing gradients, as opposed to finding conservative circular or ellipsoidal approximations to the polygons or approximating inner convex safe sets that do not always allow for planning on longer horizons. Furthermore, the interpolants can easily be buffered to account for limited accuracy in the a priori environment information. Radial Basis Functions (RBFs) [15] are used as the means for interpolating the ENC polygon data, where the choice fell on using a Thin Plate Spline (TPS) representation due to its wide usage as a non-rigid transformation model in shape matching and image alignment. It has the advantage of being a smooth and differentiable obstacle representation, making it suitable for usage as a non-linear constraint in planners or controllers utilizing gradient-based numerical optimization. It can also be used as a repulsive force field in artificial potential field methods [16]. The set of smooth surface functions representing the unstructured maritime environment is used as path constraints in a Nonlinear MPC (NMPC) trajectory

planner/tracking application for anti-grounding functionality. We show that the approximation error with the proposed static obstacle representation is not excessive in the studied examples. Lastly, we demonstrate the real-time feasibility of our approach in a coastal trajectory tracking and anti-grounding example for a ship, where the nominal trajectory to follow is initially infeasible with respect to the adjacent grounding hazards.

II. PRELIMINARIES

Vectors and matrices are written as boldface symbols. The Euclidean norm is written as $\|\mathbf{x}\|$ for a vector $\mathbf{x} \in \mathbb{R}^n$, with n denoting the vector space dimension. Similarly, a matrix is written as $\mathbf{A} \in \mathbb{R}^{n \times m}$ and has n rows and m columns, with $\mathbf{A}^T$ denoting its transpose. For square matrices where $n = m$, its inverse is given by $\mathbf{A}^{-1}$. The $n \times n$ identity matrix is written as $\mathbf{I}_n$ or $\mathbf{I}_{n \times n}$, and the same for a matrix of zeros $\mathbf{0}_n$. A positive definite matrix $\mathbf{A}$ satisfies $\mathbf{x}^T \mathbf{A} \mathbf{x} > 0$ for all non-zero $\mathbf{x}$. Coordinate frames are specified by $\{\cdot\}$. We consider polynomial functions $p : \mathbb{R}^n \rightarrow \mathbb{R}$ that are elements in the set $\mathbb{P}_n^d$ of polynomials with degree d in n variables.

III. STATIC OBSTACLE APPROXIMATION

A. Problem Formulation

For the combined trajectory tracking and anti-grounding problem, we consider an unstructured environment represented by a covering of n_{so} static obstacle polygons $j = 1, \dots, n_{so}$ obtained from an Electronic Navigational Chart (ENC). Each polygon is represented by a set or sequence of $\mathcal{P}^j = \{\zeta_i\}_{i=1}^{n_v^j}$ planar data points $\zeta_i = [x_i, y_i]^T \in \mathbb{R}^2$. The aim is to construct a set $\mathcal{H}$ of smooth and twice differentiable C^2 surface functions $\{h_j(\zeta)\}_{j=1}^{n_{so}} \in \mathcal{H}$, where each $h_j : \mathbb{R}^2 \rightarrow \mathbb{R}$ approximates the non-convex polygon j through interpolation. Granted the surface set, we identify grounding with a static obstacle j as $h_j(\zeta) > 0$ and $h_j(\zeta) \leq 0$ as no grounding. To make the interpolation problem well-posed, function values $h_j^i = h^+, \forall i = 1, \dots, n_v^j$ are assigned for the polygon point set $\mathcal{P}^j$, where $h^+ > 0$ is some positive value, typically 1. Furthermore, each surface approximation should only satisfy $h_j(\zeta) > 0$ on the actual polygon domain. Thus, the polygon point set $\mathcal{P}^j$ is expanded into $\tilde{\mathcal{P}}^j$ by a buffer point set $\{\zeta_l\}_{l=1}^{n_b^j}$ created by expanding polygon j by a buffer distance d_{buff} , which is related to the ENC accuracy as well as control accuracy requirements. At these n_b^j buffer points, we assign the negative function value $h_j^l = h^- < 0, \forall l = 1, \dots, n_b^j$. In general, d_{buff} should be small in order to prevent a conservative surface approximation. Extra buffer points far away from the polygon can also be added to $\mathcal{P}^j$ to prevent unwanted function behavior outside the polygon domain. This procedure results in an interpolation data set consisting of tuples $\{\zeta_i, h_j^i\}_{i=1}^{n_p^j}$ with $n_p^j = n_v^j + n_b^j$ as the total number of data points, for which we must fit suitable functions $h_j(\cdot)$.

B. Method

To solve the interpolation problem described above, one may use polynomial interpolants or piecewise polynomials. A high-order polynomial function is, however, not suited for interpolation. Piecewise polynomials require a division of $\mathbb{R}^2$ into a triangular or rectangular mesh, with different polynomials being applied on different patches, stitched together using boundary constraints to ensure continuity. However, for scattered data interpolation problems such as the one posed in this work, piece-wise polynomials struggle with spurious bumps and overshoots when there are large data-free regions [17]. In contrast, Radial Basis Function (RBF) interpolants can adjust automatically to varying data density and guarantee an invertible set of linear equations under very mild conditions on the data.

An RBF is a function $\phi(\|\zeta - \zeta_i\|) : \mathbb{R}_+ \rightarrow \mathbb{R}$ centered at $\zeta_i = [x_i, y_i]^T$ [15], and the RBF interpolant is typically written as

$$h_j(\zeta) = \sum_{i=1}^{n_p^j} w_i \phi(\|\zeta - \zeta_i\|) + p(\zeta) \quad (1)$$

where a low-order polynomial $p(\cdot) \in \mathbb{P}_n^{d-1}$ is often added to assure a well-posed problem. This will be explained later in this section. Popular choices of $\phi(r)$, with r as the input radial distance, include the Gaussian RBF $\phi(r) = e^{-(\epsilon r)^2}$ with $\epsilon \in \mathbb{R}$ being a shape parameter, and the Thin Plate Spline (TPS) $\phi(r) = r^2 \log(r + \delta)$, where the singularity of the logarithmic function at the origin is avoided with a small regularization parameter $\delta > 0$. The TPS achieves several times continuous differentiability, which makes it suitable for usage in MPC as constraints, where C^2 is required when using e.g. Sequential Quadratic Programming (SQP) [18]. As a result, we assume usage of RBF functions $\phi(r) : (0, \infty) \rightarrow \mathbb{R}$ away from the polygon points. Further note that, although the TPS does not require a scaling parameter, we shift the input polygon point sets by the bounds of the geographical region the vessel operates in.

The interpolation requirements are $h_j(\zeta_i^j) = h_j^i \quad \forall_i$. Then, if the interpolation data points are unique, the interpolation matrix

$$\mathbf{A}_j = \begin{bmatrix} \phi(\|\zeta_1 - \zeta_1\|) & \dots & \phi(\|\zeta_1 - \zeta_{n_p^j}\|) \\ \vdots & \ddots & \vdots \\ \phi(\|\zeta_{n_p^j} - \zeta_1\|) & \dots & \phi(\|\zeta_{n_p^j} - \zeta_{n_p^j}\|) \end{bmatrix} \in \mathbb{R}^{n_p^j \times n_p^j} \quad (2)$$

will be positive definite, making it possible to find a unique set of coefficients w_i in the interpolant. Then we say that the RBF $\phi(\cdot)$ is positive definite as well. However, in special cases such as if using the TPS, the positive definiteness is only *conditional* up to some order d . The *conditional positive definiteness* requirement is then that $\mathbf{A}_j$ is non-negative definite on the subspace of weight vectors $\mathbf{w}_j = [w_1, w_2, \dots, w_{n_p^j}]^T \in \mathbb{R}^{n_p^j}$ that are orthogonal to the polynomial space $\mathbb{P}_n^{d-1}(\bar{\mathcal{P}}^j)$, which are the set of polynomials

of degree $d - 1$ restricted to the point set $\bar{\mathcal{P}}^j$ [15]. This is written as the requirement

$$\mathbb{R}^{n_p^j} \ni \mathbf{w} \perp \mathbb{P}_n^{d-1}(\bar{\mathcal{P}}^j) \leftrightarrow \sum_{i=1}^{n_p^j} w_i q(\zeta_i) = 0, \forall q \in \mathbb{P}_n^{d-1}, \quad (3)$$

where q is a polynomial function belonging to the space $\mathbb{P}_n^{d-1}$. In our case when $d = 2$, the requirements means that the set of data points in $\bar{\mathcal{P}}^j$ must not be the subset of a straight line [19]. The requirement is naturally satisfied by most of the ENC polygon data, as the polygons normally have areas strictly larger than zero. If this is not the case, one should pre-process the data to make the problem well-posed. Considering an augmented polynomial of degree $d = 1$ and thus $p(\zeta) = c_1 + c_2 x + c_3 y$, the parameter vector $\mathbf{w}$ and polynomial coefficients are found by solving a system of equations (1) and (3)

$$\begin{bmatrix} \mathbf{A}_j + \Omega \mathbf{I}_{n_p^j} & \mathbf{B}_j \\ \mathbf{B}_j^T & \mathbf{0}_{n_{pol}} \end{bmatrix} \begin{bmatrix} \mathbf{w}_j \\ \mathbf{c}_j \end{bmatrix} = \begin{bmatrix} \mathbf{h}_j \\ \mathbf{0}_{3 \times 1} \end{bmatrix} \quad (4)$$

where

$$\mathbf{B}_j = \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_{n_p^j} & y_{n_p^j} \end{bmatrix}, \quad \mathbf{h}_j = \begin{bmatrix} h_j^{\zeta_1} \\ \vdots \\ h_j^{\zeta_{n_p^j}} \end{bmatrix}, \quad \mathbf{c}_j = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \quad (5)$$

in which $\mathbf{h}_j$ is a stacked vector of the underlying polygon interpolant function values, $\mathbf{c}_j$ the polynomial coefficient vector with $n_{pol} = 3$ as $d = 1$. A smoothing or regularization term with tuneable parameter $\Omega > 0$ has been included in the system of equations to adjust how well the interpolant fits the data. When applying the thin-plate spline to the problem, one essentially minimizes the energy functional

$$E[h] = \int_{\mathbb{R}^2} \left(\frac{\partial^2 h}{\partial x^2} \right)^2 + 2 \left(\frac{\partial^2 h}{\partial x \partial y} \right)^2 + \left(\frac{\partial^2 h}{\partial y^2} \right)^2 dx dy \quad (6)$$

for all interpolants giving a well-defined functional. In environments with mapped polygons varying greatly in size, one should consider the usage of multiscale hierarchical RBF (MHRBF) networks for increased accuracy [20]. These can self-organize, where RBF grids with units of decreasing scale are inserted at regions with local errors above a certain threshold. This is, however, not an issue in this work, and MHRBF networks will thus not be utilized here. For more information on RBFs, the interested reader is referred to e.g. [21].

C. Approximation Results

We consider the map of grounding hazards shown in Fig. 1 fetched using [22], and a provided nominal trajectory that the NMPC described in the next section will track, which is not feasible. This requires anti-grounding functionality in the tracking controller. For this purpose, only hazards inside a corridor of radius $r_{rel} = 200$ m about the nominal trajectory is considered. We buffer the hazards by a safety margin $d_{safe,so} = 5$ m, which in general depends on the

ship's size, draft, maneuverability, navigation accuracy, as well as ENC map accuracy, and environmental factors such as wave height, tides, and seabed properties. The relevant grounding hazards are found by computing the intersection between the corridor and polygons in the map, which are then approximated by TPSs using the proposed interpolation method. This results in $n_{so} = 11$ considered grounding hazards.

Results using the proposed polygon surface approximation method with $\Omega = 1.0$ for one of the grounding hazards are given in Fig. 2. To find the corresponding approximation error related to how well the surface fits the entire polygon domain on $\mathbb{R}^2$, we create a mask ground truth $\underline{h}_j(\zeta)$ for each surface. The mask satisfies $\underline{h}_j(\zeta) > 0, \zeta \in \mathcal{B}_j$ on the polygon domain $\mathcal{B}_j$, and $\underline{h}_j(\zeta) \leq 0$ otherwise. Then, data points are sampled with 0.5 m resolution in north and east coordinates on a rectangular grid, on which the ground truth mask and surface function values $h_j(\zeta)$ are calculated and compared. Fig. 3 shows the results for the polygon in Fig. 2, where $h_j(\zeta) > 0$ outside the actual polygon domain. Condensed numerical results for all relevant polygons are shown in Table I. Naturally, the error is larger in areas where the polygon point set is more scarce. Also, when expanding the polygon with a larger safety margin, the original detail level can get distorted depending on the buffering method used. This issue can be seen from increased approximation error between the two promontories in the southern part of the obstacle plot. Thus, one should use a smaller safety margin for e.g. docking applications. However, for larger ships in transit, this level of detail may be less important, thus allowing for larger safety margins to be used. Note that the ENC data accuracy in the given region should always be taken into account in the buffering.

From Table I, one can see that the proposed static obstacle representation is reasonably accurate and not overly conservative as one could get using circular or ellipsoidal approximations. The maximum error typically occurs between promontories or narrow bays, or between polygon vertices where there are no point data. Increased accuracy can here be obtained by adding more points along the polygon boundary or by reducing Ω . This will naturally be a trade-off with the computational effort in evaluating the surface function. This computational expense can also be remedied by e.g. using a set of nearest neighbors instead of the whole point set in (1) to evaluate the spline function.

Guaranteeing $h_j(\zeta) > 0$ on the polygon boundary and its interior for the TPS interpolant is hard, and will depend heavily on the data scaling and shifting [23], the smoothing parameter Ω , the polygon coverage by the point set and the distribution between these points. However, given a representative polygon point set, one can select suitably large h^+ , h^- and Ω , such that $h_j(\zeta) > 0$ on the polygon support. Note that this comes at the cost of conservatism and larger interpolant gradient magnitudes which can lead to ill-conditioned matrices and high sensitivity in the numerical optimization routines.

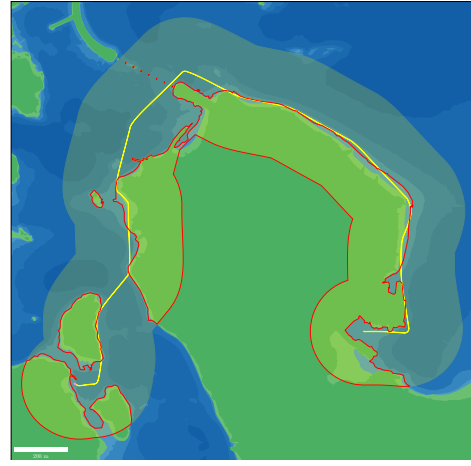


Fig. 1: Corridor (transparent yellow) around the nominal path/trajectory (yellow) to follow, with relevant grounding hazards for the MPC to consider (red). The grounding hazards include land, shore, and seabed less than or equal to the ship draft.

TABLE I: Results for all relevant polygon approximations, calculated for a grid of points with resolution 0.5 m in x and y , measuring the distance from the actual polygon to where $h_j(\zeta) > 0$ outside the actual polygon boundary (similarly to Fig. 3). Here, ϵ_{avg} is the mean error, ϵ_{max} the maximum error and σ the standard deviation.

j	n_p^j	$\epsilon_{\text{avg}} [m]$	$\epsilon_{\text{max}} [m]$	$\sigma [m]$
1	287	0.002	4.167	0.056
2	172	0.006	3.991	0.096
3	92	0.009	4.314	0.120
4	26	0.006	1.672	0.073
5	26	0.006	1.459	0.070
6	24	0.006	1.650	0.072
7	26	0.013	2.538	0.136
8	26	0.011	2.347	0.122
9	811	0.003	8.172	0.087
10	26	0.012	2.385	0.126
11	26	0.013	2.555	0.131

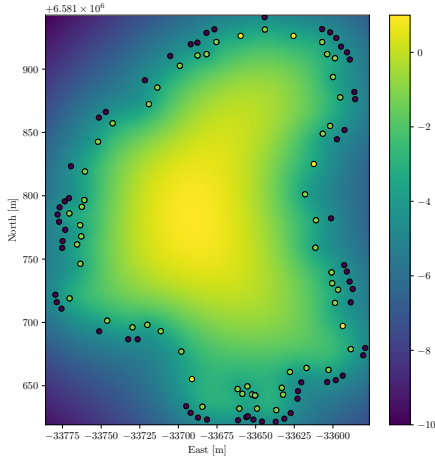
IV. TRAJECTORY TRACKING NMPC WITH ANTI-GROUNDING

A. Ship Dynamics

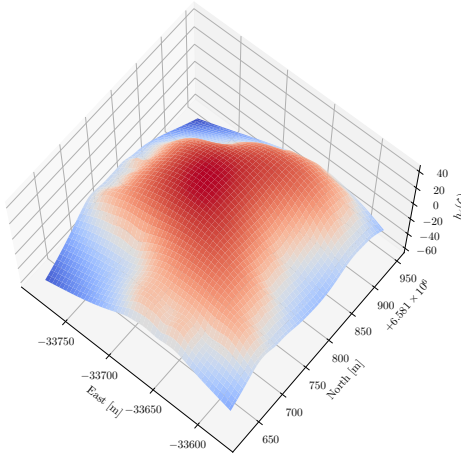
A surface vessel model with 3 degrees of freedom (DOF) is used to describe the ship dynamics in surge, sway, and yaw as in [24]. The model considers a rudder and propeller as control inputs. Let $\mathbf{x} = [\boldsymbol{\eta}^T, \boldsymbol{\nu}^T]^T \in \mathcal{X} = \mathbb{R}^{n_x}$ be the ownship state ($n_x = 6$), consisting of its pose $\boldsymbol{\eta} = [x, y, \psi]^T \in \mathbb{R}^3$ in the North-East-Down (NED) $\{n\}$ frame and body-frame $\{b\}$ velocity $\boldsymbol{\nu} = [u, v, r]^T \in \mathbb{R}^3$. The input vector is given as $\mathbf{u} = [F_x, F_y]^T \in \mathcal{U} = \mathbb{R}^{n_u}$ ($n_u = 2$), where F_x and F_y are forces applied to the vessel by the main propeller and rudder, directed along the vessel x and y axes, respectively. See Fig. 4 for an illustration. The model can be written on



(a) The considered polygon, expanded by a safety margin $d_{\text{safe},so}$ shown in black. The red boundary shows the actual grounding hazard boundary.



(b) Colormesh plot with the interpolation data set shown. The extra buffer points are shown in approximate black color, whereas the actual polygon safety margin $d_{\text{safe},so}$ buffered boundary points are in green and yellow-like colors.



(c) The RBF surface approximation.

Fig. 2: RBF approximation of a non-convex island polygon.

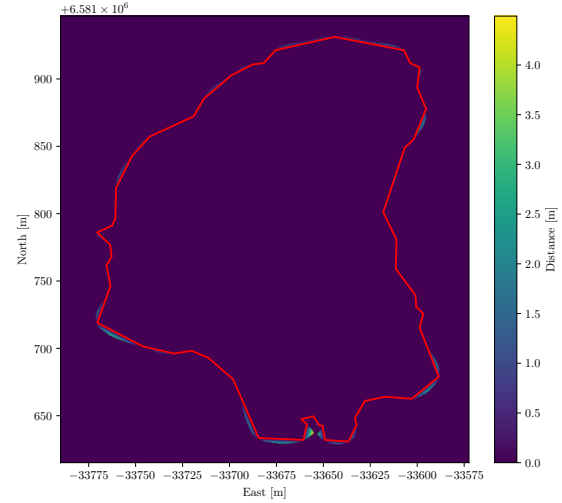


Fig. 3: RBF surface approximation error plot for the polygon in Fig. 2. The actual polygon boundary is shown in red, whereas the color bar denotes the distance outward from the polygon where the surface is still $h_j(\zeta) > 0$.

the form

$$\dot{x} = f(x, u) = \begin{bmatrix} J(\psi)\nu \\ M^{-1}(-C(\nu)\nu - D(\nu)\nu + \tau) \end{bmatrix} \quad (7)$$

which defines the dynamics $f : \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ with

$$\tau = Ku = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & -l_r \end{bmatrix} u \quad (8)$$

Here l_r is the distance from the vessel center of gravity to the rudder, along the body x -axis. The transformation matrix $J(\psi)$ is given by

$$J(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (9)$$

from the NED frame $\{n\}$ to the body frame $\{b\}$.

Further, the inertia matrix is given by $M \in \mathbb{R}^{3 \times 3}$, $C(\cdot) \in \mathbb{R}^{3 \times 3}$ is the coriolis and centripetal matrix and $D(\cdot) \in \mathbb{R}^{3 \times 3}$ the nonlinear damping matrix. For a vessel with typical port-starboard symmetry, expressions can be found in [24]. Environmental disturbance forces are not considered, as we assume small disturbances without loss of generality. Note that slowly varying disturbances can be compensated for in the NMPC by e.g. using a disturbance observer and augmenting the model with an integrator state. Furthermore, the pose and velocity of the ship are assumed to be accurately measured, and we thus neglect the corresponding navigation uncertainty.

B. Optimal Control Problem

Nonlinear Model Predictive Control (NMPC) is applied for planning a modified grounding-free trajectory that follows

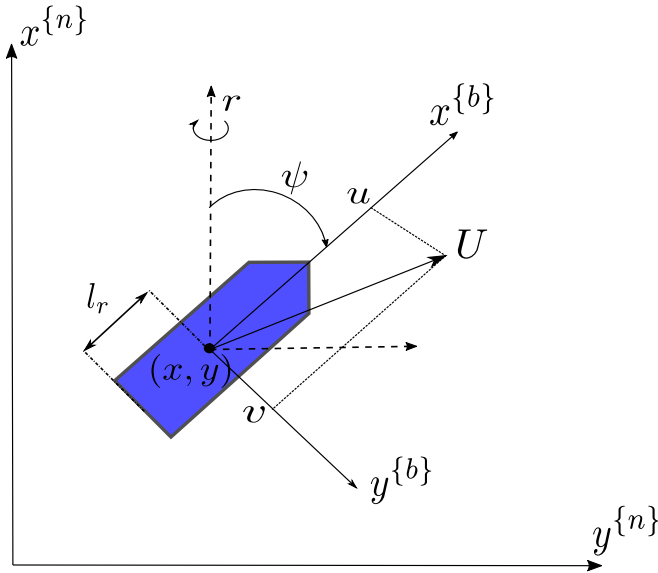


Fig. 4: Own-ship geometry with the model variables shown, with U as the ship speed. The own-ship position (x, y) is given in the NED frame and is located at the vessel center of gravity.

the nominal trajectory shown in the previous section as closely as possible, given the set static hazard safety margin. As the mapped static obstacles can not be directly used as constraints in the MPC framework, the surface function approximations found in the previous section are used in replacement directly as non-linear path constraints instead. Then, we consider the following Optimal Control Problem (OCP) underlying the proposed NMPC

$$\min_{x, u} \int_{t_0}^{t_0+T_{\text{mpc}}} \Phi(x(t), u(t)) dt + \Gamma(x(t_0 + T_{\text{mpc}})) \quad (10a)$$

$$\text{s. t. } \dot{x} = f(x(t), u(t)), \quad \forall t \in [t_0, t_0 + T_{\text{mpc}}] \quad (10b)$$

$$h(x(t)) \leq 0, \quad \forall t \in [t_0, t_0 + T_{\text{mpc}}] \quad (10c)$$

$$u_{\min} \leq u(t) \leq u_{\max}, \quad \forall t \in [t_0, t_0 + T_{\text{mpc}}] \quad (10d)$$

with t_0 as the current time and $x(t_0)$ the current time state information. A quadratic stage cost

$$\Phi(x, u) = \tilde{x}^T Q \tilde{x} + \tilde{u}^T R \tilde{u} \quad (11)$$

and terminal cost

$$\Gamma(x) = \tilde{x}^T P \tilde{x} \quad (12)$$

is used, where the continuous-time dependency is dropped for the ease of notation, and is thus implicit in e.g. $x = x(t)$. The terminal cost is used to approximate the remaining infinite horizon cost. Here, the tracking error in states and inputs are given by $\tilde{x} = x_d - x$ and $\tilde{u} = u_d - u$, respectively. Positive-definite matrices $Q \in \mathbb{R}^{n_x \times n_x}$, $R \in \mathbb{R}^{n_u \times n_u}$, and $P \in \mathbb{R}^{n_x \times n_x}$ are used as cost weights. The ship dynamics equality constraint is given by (10b). For convenience, the static obstacle safety-related inequality constraints are gathered in the vector $h(x(t)) \in \mathbb{R}^{n_{so}}$ with elements $h_j(x, y)$ in (10c), whereas (10d) gives the input constraints.

C. Nonlinear Program

To solve (10), we employ discretization using direct multiple shooting [25] on a suitably defined time mesh $t_0 < t_1 < \dots < t_k < \dots < T_{\text{mpc}}$ with sample time Δ_{mpc} and horizon T_{mpc} . Explicit Runge Kutta is used as a numerical integrator to define the discrete-time dynamics $x_{k+1} = F(x_k, u_k) \quad \forall \quad k \in 0, N-1$ with $N = T_{\text{mpc}}/\Delta_{\text{mpc}}$, which gives

$$l_1 = f(x_k, u_k) \quad (13a)$$

$$l_2 = f(x_k + \frac{\Delta_{\text{mpc}}}{2} l_1, u_k) \quad (13b)$$

$$l_3 = f(x_k + \frac{\Delta_{\text{mpc}}}{2} l_2, u_k) \quad (13c)$$

$$l_4 = f(x_k + \Delta_{\text{mpc}} l_3, u_k) \quad (13d)$$

$$F(x_k, u_k) = x_k + \frac{\Delta_{\text{mpc}}}{2} (l_1 + 2l_2 + 2l_3 + l_4) \quad (13e)$$

For each time index k , the initial state constraint and discrete-time dynamics can then be stacked into the vector of equality constraints

$$g(x_{k+1}, x_k, u_k) = \begin{bmatrix} x_0 - x(t_0) \\ x_1 - F(x_0, u_0) \\ \vdots \\ x_N - F(x_{N-1}, u_{N-1}) \end{bmatrix}, \quad (14)$$

and (10) can then be transcribed to the following nonlinear program

$$\min_{x, u} \sum_{k=0}^{N-1} (\Phi(x_k, u_k) + \lambda^T s_k) \Delta_{\text{mpc}} + \lambda^T s_N + \Gamma(x_N) \quad (15a)$$

$$\text{s.t. } x_0 = x(t_0) \quad (15b)$$

$$g(x_{k+1}, x_k, u_k) = 0, \quad \forall k \in [0, N-1] \quad (15c)$$

$$h(x_k) \leq 0 + s_k, \quad \forall k \in [0, N] \quad (15d)$$

$$u_{\min} \leq u_k \leq u_{\max}, \quad \forall k \in [0, N-1] \quad (15e)$$

$$0_{n_{so} \times 1} \leq s_k, \quad \forall k \in [0, N] \quad (15f)$$

where slack variables $s_k \in \mathbb{R}^{n_{so}}$ are used to soften the nonlinear anti-grounding path constraints to ensure feasibility, using an L_1 -norm term with weight vector $\lambda \in \mathbb{R}^{n_{so}}$ for penalizing non-zero slacks. The NMPC problem is well posed with a quadratic cost function, smooth C^2 surface splines $h_j(x, y)$ and dynamics. Thus, by using SQP with a primal-dual interior point method for the underlying QP-solver (see our choice in Section V) from an initially feasible guess with suitable cost weights, the convergence of the SQP-iterates can be locally quadratic [18][26].

For simplicity, we assume perfect tracking of the NMPC output optimal trajectory by the ship's low-level motion control system. For deployment, however, the entire pipeline with the anti-grounding planner and low-level motion controller would naturally be tuned and validated as a whole. Utilizing a disturbance estimator and/or integral action in the NMPC will allow for slowly time-varying environmental force compensation, at the cost of higher complexity. Modifying the framework to achieve this is straightforward.

V. SIMULATION RESULTS

The NMPC planner with embedded anti-grounding is used to track the nominal trajectory in Fig. 1 from Linehamna to Åmøy harbor near Stavanger, Norway. The total distance traveled is approximately 3 km, and a reference speed of 4.0 m/s is used to generate the trajectory. Acados [27] is used to formulate the OCP in (10) when using the ship model in (7), where SQP is used together with the High-Performance Interior Point Method solver [28] for computing the NMPC solutions of (15). Tuning parameters $T_{\text{mpc}} = 40.0$ s, $\Delta_{\text{mpc}} = 0.5$ s, $\mathbf{Q} = \text{diag}([0.3, 0.3, 15.0, 10.0, 7.5, 1.0])$, $\mathbf{R} = \text{diag}([0.0001, 0.0001])$, $\mathbf{P} = \mathbf{Q}$, $\lambda = 12000.0 \cdot \mathbf{1}_{n_{so} \times 1}$ are used in the NMPC. The simulation framework in [29] is used for testing the method.

Example tracking results are presented in Figs. 5 - 7. A nominal trajectory buffer and static obstacle safety margin of $r_{\text{rel}} = 200$ m and $d_{\text{safe},so} = 5.0$ m are used. Without excessive efforts being spent on tuning, one can see from the plots that the proposed anti-grounding control system onboard the own-ship is able to steer safely around static hazards with little conservatism, staying close to the non-convex $d_{\text{safe},so}$ -buffered polygon boundary while tracking the trajectory. Variations in the heading, sway, and yaw rate arise naturally due to the system tracking the nominal trajectory while circumventing the contours of the RBF grounding hazard representation. Run-time results on a PC averaged over the entire scenario gave NMPC computation time 0.24 ± 0.17 s with a maximum of 0.75 s, proving that the method is viable for deployment at the used sampling rate. Note that further reductions in run-time can be achieved by e.g. only considering static hazards inside a specific radius, reducing the number of interpolation data points, and using nearest neighbor-based evaluation in the TPS interpolant.



Fig. 5: North-east plot of the NMPC trajectory tracking. The nominal trajectory is shown in —, own-ship in — at each NMPC iteration, predicted NMPC trajectories in —, and safety buffered grounding hazard boundaries in —.

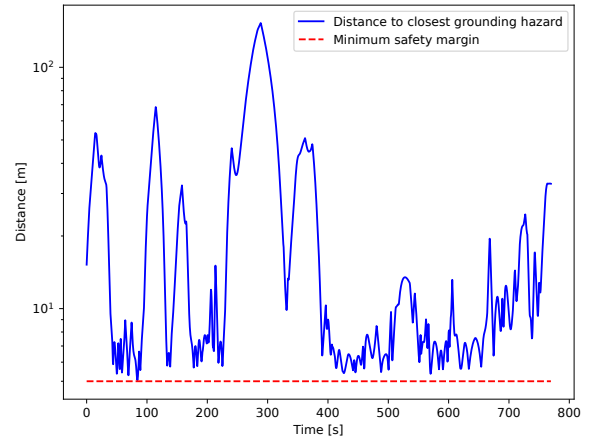


Fig. 6: Distance from the own-ship to the closest grounding hazard.

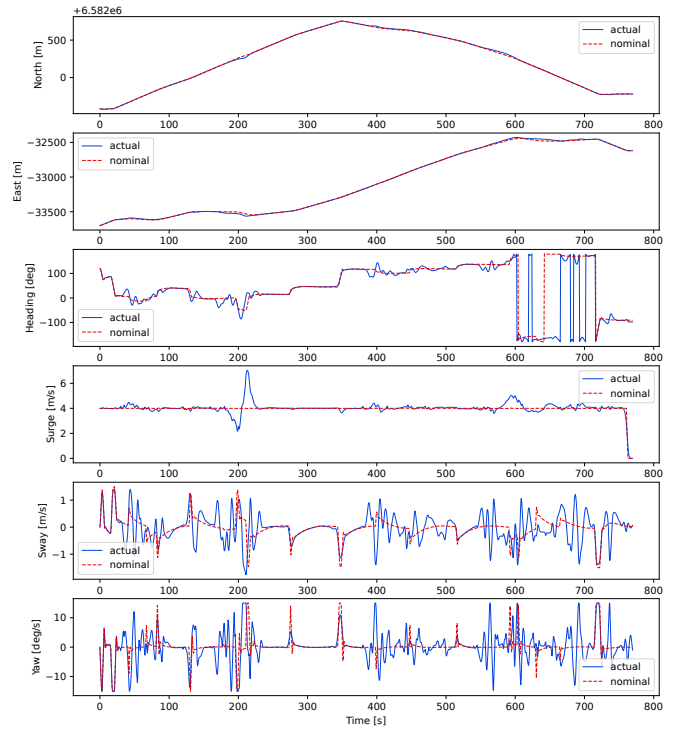


Fig. 7: NMPC tracking results.

VI. CONCLUSION

Nonlinear Model Predictive Control (NMPC) applications for anti-grounding and collision avoidance have predominantly relied on simplistic and conservative static obstacle representations. However, these approaches prove impractical for autonomous navigation in narrow channels, near-land surveys, and docking scenarios where a more detailed consideration of non-convex polygons obtained from ENC information on the unstructured environment is crucial. To address this limitation, the use of Radial Basis Function (RBF) interpolants has been proposed to approximate non-convex polygons in the vicinity of autonomous ships operating in the

unstructured environment. RBFs offer inherent smoothness, differentiability, and the ability to approximate polygons described by scattered data sets, making them advantageous for optimal control applications in anti-grounding and collision avoidance. To account for control accuracy requirements and inaccuracies in the polygons obtained from ENC data, the surface interpolants can be expanded by simply buffering the data point sets.

Interpolation results employing a TPS-type RBF demonstrate that the approximation error remains within acceptable limits while capturing the polygon details. These TPS interpolants are directly utilized as path constraints within an NMPC planner/trajectory tracker. Through simulations involving multiple non-convex grounding hazards of varying size and complexity, we establish that the anti-grounding control system successfully tracks initially non-feasible trajectories without grounding, closely adhering to the hazard boundaries within the predefined safety margin.

Future advancements will involve integrating learning-based techniques to adapt safety margins dynamically in response to external disturbances and nearby ships. Additionally, testing of the proposed method within a complete pipeline incorporating a high-level trajectory planner and a low-level motion control system is necessary. Moreover, extending the NMPC to consider dynamic obstacles and adherence to the traffic rules is a natural progression for further algorithmic improvements.

REFERENCES

- [1] T. Tengesdal, T. A. Johansen, T. D. Grande, and S. Blindheim, "Ship collision avoidance and anti grounding using parallelized cost evaluation in probabilistic scenario-based model predictive control," *IEEE Access*, vol. 10, pp. 111 650–111 664, 2022.
- [2] L. Blackmore, M. Ono, and B. C. Williams, "Chance-constrained optimal path planning with obstacles," *IEEE Transactions on Robotics*, vol. 27, no. 6, pp. 1080–1094, 2011.
- [3] S. M. LaValle, *Planning algorithms*. Cambridge university press, 2006.
- [4] S. Karaman and E. Frazzoli, "Optimal kinodynamic motion planning using incremental sampling-based methods," in *49th IEEE Conference on Decision and Control (CDC)*, 2010, pp. 7681–7687.
- [5] I. Noreen, A. Khan, and Z. Habib, "Optimal path planning using rrt* based approaches: a survey and future directions," *International Journal of Advanced Computer Science and Applications*, vol. 7, no. 11, 2016.
- [6] H. Xue, "A quasi-reflection based sc-pso for ship path planning with grounding avoidance," *Ocean Engineering*, vol. 247, p. 110772, 2022.
- [7] L. Wang, Z. Zhang, Q. Zhu, and S. Ma, "Ship route planning based on double-cycling genetic algorithm considering ship maneuverability constraint," *IEEE Access*, vol. 8, pp. 190 746–190 759, 2020.
- [8] D. E. Kirk, *Optimal control theory: an introduction*. Courier Corporation, 2004.
- [9] L. E. Dubins, "On curves of minimal length with a constraint on average curvature, and with prescribed initial and terminal positions and tangents," *American Journal of Mathematics*, vol. 79, p. 497, 1957.
- [10] H. G. Bock and K.-J. Plitt, "A multiple shooting algorithm for direct solution of optimal control problems," *IFAC Proceedings Volumes*, vol. 17, no. 2, pp. 1603–1608, 1984.
- [11] M. Abdelaal, M. Fränzle, and A. Hahn, "Nonlinear model predictive control for trajectory tracking and collision avoidance of underactuated vessels with disturbances," *Ocean Engineering*, vol. 160, pp. 168 – 180, 2018.
- [12] T. Schoels, P. Rutquist, L. Palmieri, A. Zanelli, K. O. Arras, and M. Diehl, "Ciao*: Mpc-based safe motion planning in predictable dynamic environments," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 6555–6562, 2020, 21st IFAC World Congress.
- [13] A. Vaaler, H. Robinson, T. Tengesdal, and A. Rasheed, "Safety filter for small passenger ferry," in *42nd International Conference on Ocean, Offshore & Arctic Engineering*, 2023.
- [14] A. B. Martinsen, A. M. Lekkas, and S. Gros, "Optimal model-based trajectory planning with static polygonal constraints," *IEEE Transactions on Control Systems Technology*, vol. 30, no. 3, pp. 1159–1170, 2022.
- [15] M. D. Buhmann, "Radial basis functions," *Acta Numerica*, vol. 9, p. 1–38, 2000.
- [16] Y. Xue, B. Lee, and D. Han, "Automatic collision avoidance of ships," *Proceedings of the Institution of Mechanical Engineers, Part M: Journal of Engineering for the Maritime Environment*, vol. 223, no. 1, pp. 33–46, 2009.
- [17] S. D. Billings, R. K. Beatson, and G. N. Newsam, "Interpolation of geophysical data using continuous global surfaces," *GEOPHYSICS*, vol. 67, no. 6, pp. 1810–1822, 2002.
- [18] J. Nocedal and S. Wright, *Numerical optimization*. Springer Science & Business Media, 2006.
- [19] M. J. D. Powell, "The theory of radial basis function approximation in 1990," *Advances in numerical analysis*, vol. 2, pp. 105–210, 1992.
- [20] S. Ferrari, M. Maggioni, and N. A. Borghese, "Multiscale approximation with hierarchical radial basis functions networks," *IEEE Transactions on Neural Networks*, vol. 15, no. 1, pp. 178–188, 2004.
- [21] W. A. Light, *Some Aspects of Radial Basis Function Approximation*. Dordrecht: Springer Netherlands, 1992, pp. 163–190.
- [22] S. Blindheim and T. A. Johansen, "Electronic navigational charts for visualization, simulation, and autonomous ship control," *IEEE Access*, vol. 10, pp. 3716–3737, 2022.
- [23] S. Rippa, "An algorithm for selecting a good value for the parameter c in radial basis function interpolation," *Advances in Computational Mathematics*, vol. 11, pp. 193–210, 1999.
- [24] T. I. Fossen, *Handbook of marine craft hydrodynamics and motion control*. John Wiley & Sons, 2011.
- [25] P. Deuffhard, "A modified newton method for the solution of ill-conditioned systems of nonlinear equations with application to multiple shooting," *Numerische Mathematik*, vol. 22, no. 4, pp. 289–315, 1974.
- [26] P. E. Gill and E. Wong, "Sequential quadratic programming methods," in *Mixed integer nonlinear programming*. Springer, 2011, pp. 147–224.
- [27] R. Verschuere, G. Frison, D. Kouzoupis, J. Frey, N. v. Duijken, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, "acados—a modular open-source framework for fast embedded optimal control," *Mathematical Programming Computation*, vol. 14, no. 1, pp. 147–183, 2022.
- [28] G. Frison and M. Diehl, "Hpipm: a high-performance quadratic programming framework for model predictive control," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 6563–6569, 2020.
- [29] T. Tengesdal and T. A. Johansen, "Simulation framework and software environment for evaluating automatic ship collision avoidance algorithms," *7th IEEE Conference on Control Technology and Applications (CCTA)*, 2023.